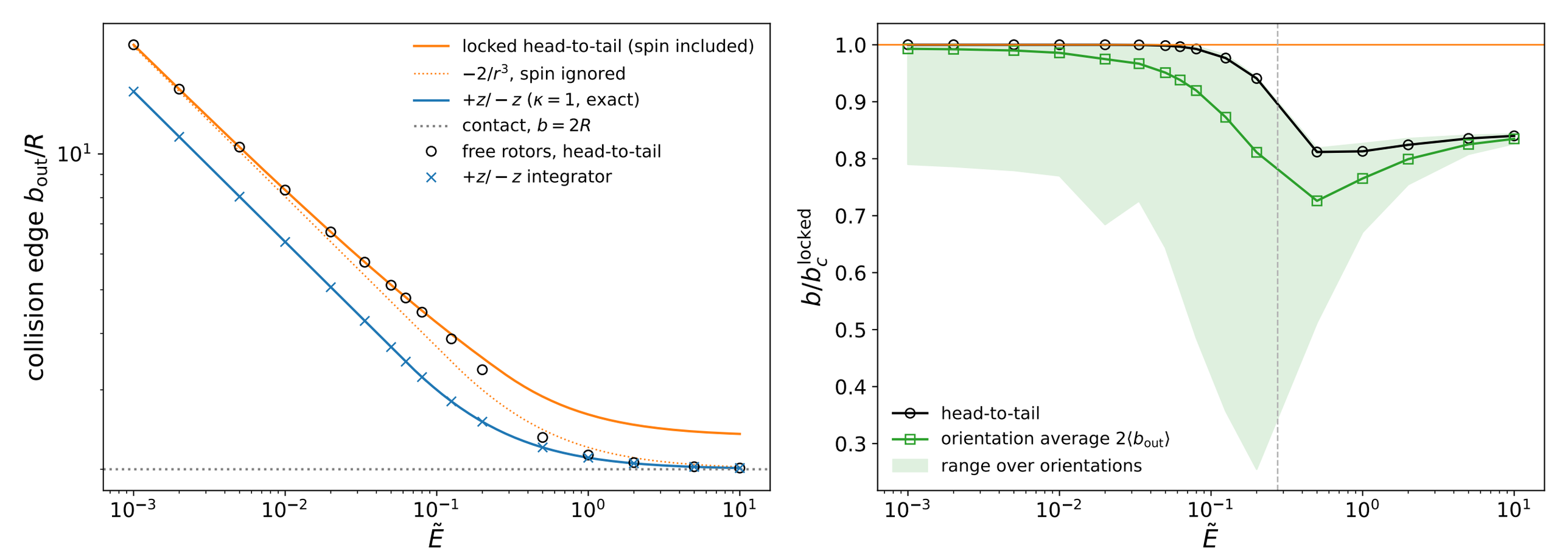
collision edge b_{out}/R
10^1
locked head-to-tail (spin included)
$-2/r^3$, spin ignored
$+z/-z$ ($\kappa = 1$, exact)
contact, $b = 2R$
free rotors, head-to-tail
$+z/-z$ integrator
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
$\tilde{E}$
b/b_c^{locked}
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
head-to-tail
orientation average $2\langle b_{\mathrm{out}}\rangle$
range over orientations
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
$\tilde{E}$